

GUI

Usage

Using GUI should be quite intuitive, just change the parameters you want and observe the difference.

Known issues

- The simulation and your whole computer might freeze for high number of neurons and low timesteps (high resolution), depending on your computer performance (mostly size of RAM). Hence, change these parameters with care.
- Changing parameters while the simulation is being evaluated will be ignored (another simulation will be run after the current one is done, but the change in parameter will be not detected, or an erroneous value will be loaded).
- Attempt was made to solve this using the NO GO button which „breaks“ the simulation call, but this did not help (all changes in parameters end up being ignored while the NO GO button is active).
- In most cases one can input negative values where only positive values make sense, so entering „random“ values can create issues.
- It is only possible to load ECAP data formatted in the manner as in Hess 2017 semester project (ECAP_time, ECAP_amplitude)

Possible modifications

- Make the code NOT run each time a parameter is changed, but only when the GO button is pressed. This will be quite important when one wants to change multiple parameters between two simulations while considering high number of neurons and iteration steps.
- Enable loading of some standardized ECAP data format (if such a thing exists), recovery and facilitation function (for various masker-pulse intensities).
- Enable loading the measured spread functions directly (so instead of fitting the spread function and loading the parameter, the whole function is loaded). If one loads the functions measured at various electrodes, this would require interpolating the function values for „intermediate“ electrodes at which no measurements were performed.
- Possibly removing the spike histogram and either adding another graph or making the current graphs bigger.

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